

Literature Handbook

Everything to know before writing Chapter 2 of ‘This Time Is Different?’

Study notes: 47 paper cards covering about 55 sources, concept primers, key numbers, Irish crisis timeline and a note-to-paragraph method · 24 September 2026

Read this first: These are study notes written in my own words from the sources, with the facts and numbers each paper reports. They contain no copied paragraphs, and they are not dissertation text. Two reasons: the papers are copyrighted, and pasting source paragraphs (or these notes) into your dissertation would be plagiarism. Use the notes to understand each paper and decide where it fits, then read the original (links in Part F) and write your own sentences, citing the original, never this handbook.

Colour code on every paper card:

Basis	Meaning
FULL	Full text read in this session — facts checked against the paper
ABSTRACT	Abstract / official summary read in this session — read the full paper for details
STANDARD	Standard textbook knowledge, not re-read this session — confirm every detail before citing

Structure: A. Key numbers · B. Irish crisis timeline · C. Concept primers · D. Paper cards (by Chapter 2 section) · E. From notes to your own paragraph · F. Where to read the originals.

A. Key numbers to remember

Numbers marked ‘your data’ come from your own pipeline (Yahoo ^ISEQ closes) — they are your results and can be reported as such in Chapter 4, with the source stated.

Fact	Number	Source
ISEQ pre-GFC peak → trough	9,981 (20 Feb 2007) → 1,916 (9 Mar 2009): -80.8%	Your data
ISEQ on 29 Sep 2008 (day before the bank guarantee)	-13.0%	Your data
ISEQ on 30 Sep 2008 (guarantee announced)	+7.9%	Your data
Irish bank CDS on guarantee day	fell overnight from ~400 to ~150 basis points while the Irish sovereign CDS rose sharply	Acharya et al. (2014)
European sample average CDS, bank / sovereign	pre-bailout 63 / 14 bp; bailout period 148 / 48 bp; post-bailout 184 / 112 bp	Acharya et al. (2014)
Irish mortgage lending, 1997 → 2008	rose to about 7 times its 1997 value	Kelly (2009)
Irish lending to developers, 1997 → 2008	about 11 times its 1997 value	Kelly (2009)
EU/IMF programme for Ireland (28 Nov 2010)	€85bn in total; €67.5bn external (€45bn EU and bilateral, €22.5bn IMF)	European Commission / press reports
Ireland exits the programme	15 December 2013	European Commission / press reports
Brexit: ISEQ on 24 and 27 Jun 2016	-7.7% and -9.9%	Your data
COVID: ISEQ on 12 Mar 2020	-9.9%	Your data
War: ISEQ on 24 Feb 2022	-4.6%	Your data
US daily moves > 2.5%, 1900–2019	more than 1,100; none attributed by newspapers to a pandemic	Baker et al. (2020)
COVID-related jumps, 24 Feb – 20 Apr 2020	about two dozen	Baker et al. (2020)
COVID excess mortality vs Spanish Flu	about 1/25th	Baker et al. (2020)
Countries in Bekaert et al.	55 equity markets	Bekaert et al. (2014)
Markets in Diebold & Yilmaz spillover study	19 global equity markets, early 1990s–2007	Diebold & Yilmaz (2009)
Christensen et al. sample	DJIA stocks, 29 Jan 2001 – 31 Dec 2017, 4,257 trading days	Christensen et al. (2023)
Bucci sample	monthly S&P 500, Feb 1950 – Dec 2017, 815 observations	Bucci (2020)

ISEQ 20 value that left in 2023–24	CRH, Flutter, Smurfit $\approx$ €74bn of $\approx$ €152bn	Irish Times
Your main sample	5,243 complete trading days, Jan 2003 – Aug 2023	Your data

B. Irish crisis timeline (2003–2023)

Dates confirmed in this session are marked ✓. Use official sources (Honohan, Regling & Watson, Nyberg, European Commission) when you cite them, not newspapers.

Date	Event	Why it matters for you
2003–2007	Irish credit and property boom; mortgage lending $\sim$ 7 $\times$, developer lending $\sim$ 11 $\times$ 1997 levels ✓	Minsky's calm-boom phase; this is your training history before the GFC
20 Feb 2007	ISEQ peak 9,981 ✓(your data)	Start of the Irish decline, before the global crisis
9 Aug 2007	BNP Paribas freezes funds; interbank markets seize	Start of your GFC window
15 Sep 2008	Lehman Brothers fails	Peak global stress; VIX spikes
30 Sep 2008	Irish government guarantees the deposits of its six biggest banks ✓	Bank risk moves onto the state: the bank–sovereign loop begins
Jan 2009	Anglo Irish Bank nationalised; state capital into AIB and Bank of Ireland ✓	Domestic banking collapse
9 Mar 2009	ISEQ trough 1,916 ✓(your data); S&P 500 trough	End of your GFC window
2009	NAMA set up to take property loans from banks	Fiscal cost of bank rescue grows
23 Apr 2010	Greece requests EU/IMF assistance	Start of your Irish sovereign-debt window
28 Nov 2010	€85bn EU/IMF programme for Ireland announced ✓	Sovereign crisis peak for Ireland
26 Jul 2012	ECB President Draghi: 'whatever it takes'	End of your sovereign-debt window
15 Dec 2013	Ireland exits the programme ✓	Recovery (Whelan: turnaround from 2012)
23–27 Jun 2016	Brexit referendum; ISEQ -7.7% and -9.9% ✓(your data)	UK-origin validation event
Feb–Mar 2020	COVID-19; ISEQ -9.9% on 12 Mar ✓(your data)	Exogenous crisis window
24 Feb 2022	Russia invades Ukraine; ISEQ -4.6% ✓(your data); ECB starts raising rates from July 2022	European energy/rates validation event
Sep 2023 – Jul 2024	CRH, Flutter, Smurfit leave the ISEQ ✓	Your sample ends 31 Aug 2023

C. Concept primers (in plain words)

C1. Crisis theory

- **Efficient Market Hypothesis (Fama, 1970).** Weak form: past prices cannot predict future returns. Semi-strong: all public information is already in prices. Strong: even private information is. Key point for you: the EMH is about the direction of returns. Volatility (how big moves will be) clusters and is forecastable without contradicting the EMH. This is why your target is volatility, not price.
- **Minsky's Financial Instability Hypothesis (1992; full text checked).** Three kinds of borrower: hedge units can pay interest and principal from income; speculative units can pay interest but must roll over principal; Ponzi units cannot even pay interest and rely on rising asset prices. Theorem 1: an economy has financing regimes in which it is stable and others in which it is unstable. Theorem 2: during long periods of prosperity, the economy moves from stable to unstable financing. Minsky also notes that if authorities fight inflation with tight money when many units are speculative, those units turn Ponzi and net worth evaporates — relevant to your 2022 rate shock.
- **Kindleberger's anatomy of a crisis (Kindleberger & Aliber, 2015).** Displacement (a new opportunity) → boom fuelled by credit → euphoria and over-trading → distress as insiders sell → revulsion and panic → lender of last resort. Use it to describe how each of your crises started.

- **‘This time is different’ syndrome (Reinhart & Rogoff).** Because big crises are years or decades apart, each generation believes the old rules no longer apply. Your RQ3 tests the machine-learning version: does a model trained on past crises still work in the next one?

C2. Contagion and transmission

- **Interdependence vs contagion.** Interdependence = normal, stable links between markets. Contagion = a significant increase in those links during a crisis (Forbes & Rigobon, 2002).
- **The correlation bias (Forbes & Rigobon; full text checked).** Correlation between markets rises mechanically when volatility rises, so naive tests find ‘contagion’ in every crisis. Their volatility-adjusted correlation is $\rho^* = \rho / \sqrt{1 + \delta(1 - \rho^2)}$, where δ is the relative increase in the variance of the source market. After adjustment they find virtually no contagion in 1987, 1994 and 1997 — only interdependence. Lesson for you: a rising SHAP share in a crisis could partly be a volatility effect. Working with shares (not raw sizes) and comparing across crises of different origin is your partial answer; say so.
- **Heat waves vs meteor showers (Engle, Ito & Lin, 1990; checked).** ‘Heat wave’: volatility driven by a market’s own news, persisting locally. ‘Meteor shower’: volatility travelling from one market to the next as they open. They rejected the pure heat-wave view: spillovers exist. Your ‘domestic’ channel is the heat wave; your six foreign channels are meteor showers.
- **Spillover index (Diebold & Yilmaz, 2009; checked).** Fit a VAR to several markets; the spillover index is the share of each market’s forecast-error variance caused by shocks from other markets. Finding: return spillovers trend slowly upward with no bursts; volatility spillovers show no trend but clear bursts in crises.
- **Wake-up call hypothesis (Bekaert et al.).** In a crisis investors re-examine each country’s own fundamentals, so weak countries are punished even without direct links to the crisis source.

C3. Volatility econometrics

- **Log return.** $r(t) = \ln(P(t)/P(t-1))$. Returns add up over time; prices do not.
- **Stylised facts (Cont, 2001).** Returns show little autocorrelation, but squared/absolute returns are strongly autocorrelated (volatility clustering) with slow decay (long memory); fat tails; losses larger than gains (asymmetry); volatility rises after falls (leverage effect).
- **GARCH(1,1).** Today’s variance $\sigma^2(t) = \omega + \alpha \cdot \varepsilon^2(t-1) + \beta \cdot \sigma^2(t-1)$. α = reaction to yesterday’s shock; β = memory of yesterday’s variance; $\alpha + \beta$ = persistence (usually close to 1 for equities).
- **Realised variance.** The sum of squared returns over a period — a noisy measure of true variance. Your target: $RV5(t) = r^2(t+1) + \dots + r^2(t+5)$.
- **HAR (Corsi, 2009).** $RV(t+1) = \beta_0 + \beta_d \cdot RV(\text{daily}) + \beta_w \cdot RV(\text{weekly average}) + \beta_m \cdot RV(\text{monthly average}) + \text{error}$. The idea (the ‘heterogeneous market hypothesis’): traders with daily, weekly and monthly horizons create volatility at different speeds. Estimated by OLS; it mimics long memory with just three terms.
- **QLIKE loss (Patton, 2011; checked).** $QLIKE = RV/F - \ln(RV/F) - 1$, where F is the forecast. It penalises under-prediction of volatility more than over-prediction, and — with MSE — it is one of the loss functions that ranks forecasts correctly even when RV is noisy.
- **Diebold–Mariano test.** Take the difference in losses between two models each day, $d(t)$. Test whether its average is zero, using a standard error robust to autocorrelation (Newey–West). With 5-day overlapping forecasts use $h - 1 = 4$ lags. The Harvey–Leybourne–Newbold version corrects the statistic for small samples and uses t-distribution critical values.

C4. LSTM

- **Why not a plain RNN?** Gradients shrink (vanish) or explode as they pass back through many time steps, so plain RNNs forget long history.

- **The LSTM fix (Hochreiter & Schmidhuber, 1997).** A cell state that carries information forward almost unchanged (the ‘constant error carousel’), controlled by gates. Input gate: what new information to store. Output gate: what to reveal. Forget gate: what to discard — note this gate was added later by Gers, Schmidhuber & Cummins (2000), a detail examiners like.
- **Why it suits volatility.** Volatility has long memory and non-linear responses (e.g., to large falls); gated memory can learn both. Bucci (2020) and Christensen et al. (2023) attribute neural-network gains to capturing persistence/long memory.

C5. Explainable AI

- **Shapley value.** From cooperative game theory: the fair share of a team’s payoff for each player, averaging each player’s marginal contribution over all possible orders of joining.
- **SHAP (Lundberg & Lee, 2017; checked).** Treat input features as players and the prediction as the payoff. SHAP is the unique additive attribution with three properties: local accuracy (attributions add up to the prediction minus the baseline), missingness (absent features get zero), consistency (if a feature matters more in a new model, its attribution does not fall). Estimators: KernelSHAP (any model, slow), DeepSHAP (neural networks, based on DeepLIFT), TreeSHAP (trees).
- **Integrated Gradients (Sundararajan et al., 2017; checked).** Two axioms: Sensitivity (a feature that changes the output must get non-zero credit) and Implementation Invariance (two networks that compute the same function get the same attributions). IG adds up gradients along a straight line from a baseline input to the actual input; attributions sum to $F(\text{input}) - F(\text{baseline})$ (completeness).
- **Expected gradients / GradientSHAP (Erion et al., 2021; checked).** IG averaged over many baselines drawn from the data — this is what Captum’s GradientShap approximates, and why your baseline set is a sample of real training windows.
- **Limits.** Correlated features make standard SHAP assume unrealistic combinations and give ‘very wrong’ explanations (Aas et al., 2021). Shapley-based importance has mathematical problems that need causal reasoning to fix (Kumar et al., 2020). For high-stakes decisions, interpretable models may be better than explained black boxes (Rudin, 2019).

D. Paper cards

Cards follow the order of your Chapter 2. ‘Feeds’ tells you the section and hypothesis the paper supports.

D1. Section 2.2 — Why crises recur

Fama, E. (1970) Efficient capital markets: a review of theory and empirical work. <i>Journal of Finance</i> , 25(2), 383–417.	
Source basis	Standard textbook knowledge, not re-read this session — confirm every detail before citing
Question	Do security prices fully reflect available information?
Data	Review of the empirical literature up to 1970.
Method	Classifies tests into weak, semi-strong and strong form.
Key findings	• Evidence broadly supports weak and semi-strong efficiency; little support for the strong form. • Any test of efficiency is also a test of the model of expected returns (the ‘joint hypothesis’ problem).
Limitations	About returns, not volatility; written before modern evidence on anomalies and volatility clustering.
How to use it	Open 2.2 with the efficiency baseline, then pivot: volatility is forecastable without violating efficiency. This also explains why your study forecasts risk rather than prices.
Feeds	2.2; justifies the target variable.

Minsky, H. (1992) The Financial Instability Hypothesis. <i>Levy Economics Institute Working Paper 74</i> .	
Source basis	Full text read in this session — facts checked against the paper
Question	Why do capitalist economies generate their own financial crises?
Data	Theoretical essay (builds on Keynes, Schumpeter, Fisher).
Method	Verbal theory of financing structures.

Key findings	• Hedge, speculative and Ponzi units defined by whether income covers interest and principal. • Theorem 1: stable and unstable financing regimes both exist. • Theorem 2: prolonged prosperity moves the economy from stable to unstable financing. • The more speculative and Ponzi finance, the more the system amplifies deviations rather than containing them. • Monetary tightening against inflation can turn speculative units into Ponzi units.
Limitations	No formal test; the theory is qualitative and hard to measure directly.
How to use it	Theoretical backbone of RQ3: a model trained only on the calm 2003–07 boom should fail when instability arrives. Link to Kelly (2009) for Irish evidence of the boom, and to 2022 for the tightening point.
Feeds	2.2; H3; Discussion (history lens).

Reinhart, C. & Rogoff, K. (2008/2009) This Time Is Different: A Panoramic View of Eight Centuries of Financial Crises (NBER WP 13882; book 2009).	
Source basis	Full text read in this session — facts checked against the paper
Question	What patterns appear when financial crises are studied across centuries and countries?
Data	New database from England's 14th-century default to the US sub-prime crisis; all regions, including India and China.
Method	Descriptive historical statistics.
Key findings	• Serial default is nearly universal as countries move from emerging to advanced. • Major defaults are spaced years or decades apart, creating the illusion that 'this time is different'. • Example of the syndrome: the false belief that domestic debt is a new phenomenon. • Crises often start in financial centres and spread through interest-rate shocks and commodity-price collapses. • The US sub-prime crisis was hardly unique.
Limitations	Macro and mostly annual data; about crisis occurrence, not market-level transmission or forecasting.
How to use it	Source of your title and framing; use 'crises start in financial centres' to motivate the US-origin expectation for the GFC (H2a).
Feeds	2.2, 2.3; title; H2a; H3.

Kindleberger, C. & Aliber, R. (2015) Manias, Panics and Crashes. 7th edn.	
Source basis	Standard textbook knowledge, not re-read this session — confirm every detail before citing
Question	What is the common anatomy of financial crises?
Data	Historical cases over several centuries.
Method	Narrative history built on Minsky's model.
Key findings	• Stages: displacement → credit-fuelled boom → euphoria → distress → panic → lender of last resort. • Manias spread internationally through trade, capital flows and psychology.
Limitations	Narrative; selection of cases.
How to use it	Use the stages to classify how each of your three crises began (its 'displacement').
Feeds	2.2; the origin table in 2.3.

Kelly, M. (2009) The Irish credit bubble. UCD Centre for Economic Research WP09/32.	
Source basis	Abstract / official summary read in this session — read the full paper for details
Question	How large was the Irish credit bubble, and what losses would it leave?
Data	Irish bank lending and property data to 2008–09.
Method	Descriptive analysis of lending growth and loss scenarios.
Key findings	• Mortgage lending rose to about seven times its 1997 value by 2008. • Lending to developers rose to about eleven times its 1997 value. • (Read the paper for bank funding and loss estimates.)
Limitations	Working paper; written during the crisis.
How to use it	Irish evidence of Minsky's boom; supports a strong domestic channel during the GFC (H2a).
Feeds	2.2 bridge, 2.3.2; H2a; H3.

Cont, R. (2001) Empirical properties of asset returns: stylized facts and statistical issues. Quantitative Finance, 1(2), 223–236.	
Source basis	Standard textbook knowledge, not re-read this session — confirm every detail before citing
Question	Which statistical properties are common to asset returns across markets?
Data	Many assets and markets.
Method	Survey of empirical regularities.
Key findings	• Little linear autocorrelation in returns; strong, slowly decaying autocorrelation in absolute/squared returns. • Heavy tails; gain/loss asymmetry; volatility clustering; leverage effect; volume–volatility correlation.
Limitations	Descriptive; does not propose a model.
How to use it	Explains why volatility is forecastable and why you model log RV with Student-t GARCH.

Feeds	2.2, 2.5; Chapter 3 target choice.
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D2. Section 2.3 — Three crises and Ireland

Brunnermeier, M. (2009) Deciphering the liquidity and credit crunch 2007–2008. <i>Journal of Economic Perspectives</i> , 23(1), 77–100.	
Source basis	Abstract / official summary read in this session — read the full paper for details
Question	How did relatively modest mortgage losses become a global financial crisis?
Data	Events and market indicators 2007–08.
Method	Narrative with an economic framework of amplification.
Key findings	• Mortgage write-downs of hundreds of billions; US stock market wealth fell by about \$8 trillion between October 2007 and October 2008. • Amplification mechanisms (read the paper for detail): balance-sheet ‘loss’ and ‘margin’ spirals, banks hoarding liquidity, runs on institutions, network effects.
Limitations	Written during the crisis; US-centred.
How to use it	Establish the GFC’s US credit origin and amplification; motivates US and fear (VIX) channels (H2a).
Feeds	2.3.2; H2a.

Bekaert, G., Ehrmann, M., Fratzscher, M. & Mehl, A. (2014) The global crisis and equity market contagion. <i>Journal of Finance</i> , 69(6), 2597–2649 (ECB WP 1381, 2011).	
Source basis	Full text read in this session — facts checked against the paper
Question	How was the 2007–09 crisis transmitted to equity markets worldwide?
Data	Country-industry equity portfolios in 55 markets.
Method	Factor model predicting crisis returns; contagion = unexplained increases in factor loadings and residual correlations.
Key findings	• Despite its US origin, little systematic contagion from US markets. • Systematic contagion from domestic equity markets to domestic portfolios. • Countries with weak fundamentals, poor policies and weak institutions suffered more contagion and larger falls. • Good policies — including debt and deposit guarantees during the crisis — partly shielded domestic equities. • Supports the ‘wake-up call’ hypothesis: investors refocused on country-specific characteristics.
Limitations	Definition of contagion depends on the factor model; not a forecasting study.
How to use it	Central for H2a: expect Ireland’s domestic channel to be large in the GFC, not only US channels. The guarantee point connects to the ISEQ’s +7.9% on 30 Sep 2008 in your own data.
Feeds	2.3.2, 2.4; H2a.

Acharya, V., Drechsler, I. & Schnabl, P. (2014) A Pyrrhic victory? Bank bailouts and sovereign credit risk. <i>Journal of Finance</i> , 69(6), 2689–2739.	
Source basis	Full text read in this session — facts checked against the paper
Question	Do bank bailouts transfer risk to governments, creating a two-way loop?
Data	CDS spreads on European sovereigns and banks, 2007 onwards.
Method	Theoretical model of the loop plus event study and panel regressions.
Key findings	• Ireland guaranteed the deposits of its six biggest banks on 30 Sep 2008. • Irish bank CDS fell overnight from about 400 to 150 bp while the Irish sovereign CDS rose sharply. • By autumn 2010 the Irish–German bond spread widened sharply, ending in a bailout of the Irish state. • Across Europe, average bank/sovereign CDS rose from 63/14 bp before bailouts to 148/48 bp during and 184/112 bp after. • After bailouts, changes in sovereign CDS help explain changes in bank CDS: the loop runs both ways.
Limitations	Credit markets, not equity volatility.
How to use it	Explains how the GFC turned into the Irish sovereign crisis — the historical link between your two windows — and why the domestic channel should matter in 2010–12 (H2b).
Feeds	2.3.2–2.3.3; H2b.

Honohan, P. (2010); Regling, K. & Watson, M. (2010); Nyberg, P. (2011) — the three official Irish reports.	
Source basis	Standard textbook knowledge, not re-read this session — confirm every detail before citing
Question	Why did the Irish banking system fail?
Data	Official investigations into Irish banks, regulators and policy 2003–08.
Method	Official inquiries (documents, interviews).
Key findings	• Heavy concentration of lending in property; reliance on wholesale funding from abroad. • Light-touch regulation and failures of macro-prudential policy (Honohan).

	• The crisis was largely home-grown; the global crisis was the trigger (Regling & Watson). • 'Herdning' by banks and authorities, all misjudging the same risk (Nyberg).
Limitations	Official reports, not academic tests; confirm exact wording and pages before citing.
How to use it	Primary historical sources for the domestic dimension of the GFC in Ireland (H2a) and your Discussion.
Feeds	2.3.2; H2a; Discussion.

Lane, P. (2012) The European sovereign debt crisis. Journal of Economic Perspectives, 26(3), 49–68.	
Source basis	Abstract / official summary read in this session — read the full paper for details
Question	What caused the euro-area sovereign debt crisis?
Data	Euro-area macro and fiscal data.
Method	Economic analysis and review.
Key findings	• Root cause: the flawed original design of the euro and a poor understanding of how fragile a monetary union is in a crisis. • No banking union and no euro-level buffers; errors in crisis management. • (Read for: pre-crisis capital flows and credit booms in the periphery, the Irish case, policy responses.)
Limitations	Written mid-crisis (2012).
How to use it	Frames the euro-area origin of the 2010–12 window (H2b). Lane later became Governor of the Central Bank of Ireland and ECB chief economist — useful context.
Feeds	2.3.3; H2b.

Whelan, K. (2014) Ireland's economic crisis: the good, the bad and the ugly. Journal of Macroeconomics, 39(B), 424–440.	
Source basis	Abstract / official summary read in this session — read the full paper for details
Question	What went right and wrong in Ireland's macroeconomy over the decade?
Data	Irish macro data, 2000s–2013.
Method	Economic review.
Key findings	• Mid-2000s demand rose well above potential, driven by a property bubble. • Either appropriate fiscal policy or appropriate financial regulation could have prevented the crisis. • Crisis management after 2008 and the turnaround from 2012 were reasonably successful.
Limitations	Macro focus; little on equity markets.
How to use it	Irish narrative; supports ending your sovereign window in 2012 (turnaround).
Feeds	2.3.3.

Corbet, S. & Twomey, C. (2015) European equity market contagion: an empirical application to Ireland's sovereign debt crisis. European Financial and Accounting Journal, 2015(3), 15–34.	
Source basis	Abstract / official summary read in this session — read the full paper for details
Question	Did the collapse of Irish equities spill over to European equity markets?
Data	Daily European equity returns; Irish crisis 2007–10 and the period after troika intervention.
Method	DCC multivariate GARCH (time-varying correlations).
Key findings	• Strong contagion between Irish and European equity markets during 2007–10. • Contagion eased dramatically after the troika intervention. • Bailouts/intervention were effective in limiting spillovers.
Limitations	Correlation-based (see Forbes & Rigobon); does not separate channels by origin; no forecasting.
How to use it	The most important Irish study for you. Compare: does your euro-core share fall after November 2010, as their finding would suggest?
Feeds	2.3.3, 2.4; H2b.

Baker, S., Bloom, N., Davis, S., Kost, K., Sammon, M. & Viratyosin, T. (2020) The unprecedented stock market reaction to COVID-19. Review of Asset Pricing Studies, 10(4), 742–758 (NBER WP 26945).	
Source basis	Full text read in this session — facts checked against the paper
Question	Why did COVID-19 move stock markets far more than earlier pandemics?
Data	US daily market moves since 1900; volatility since 1985; newspaper explanations of big moves.
Method	Text-based classification of next-day newspaper explanations for daily jumps.
Key findings	• More than 1,100 US daily moves above 2.5% in 1900–2019; newspapers attributed none to pandemics. • About two dozen such jumps between 24 Feb and 20 Apr 2020 were attributed to COVID. • COVID excess mortality was about 1/25th of the Spanish Flu's; 1957–58 and 1968 pandemics had about three times COVID's mortality yet left little market trace. • Best explanation: government restrictions and voluntary social distancing in a service-based economy; little support for faster information diffusion or supply-chain explanations.
Limitations	US only; early 2020 evidence.

How to use it	Establishes COVID as a genuinely new, policy-driven, exogenous shock — so a model trained on earlier crises has never seen anything like it (H3) and all channels should move together (H2c).
Feeds	2.3.4; H2c; H3; Discussion (was this time really different?).

Akhtaruzzaman, M., Boubaker, S. & Sensoy, A. (2021) Financial contagion during COVID-19 crisis. Finance Research Letters, 38, 101604.	
Source basis	Full text read in this session — facts checked against the paper
Question	How did COVID-19 contagion travel between China and the G7?
Data	Financial and non-financial firms' stock returns, China and G7, early 2020.
Method	DCC-GARCH correlations; spillover index; hedge ratios.
Key findings	• Correlations rose significantly for both financial and non-financial firms. • The rise was much larger for financial firms — they were key transmitters. • China and Japan were net transmitters of spillovers; hedging costs rose.
Limitations	Short window; no Ireland.
How to use it	Supports H2c (broad, simultaneous rise) and the role of financial markets as transmitters.
Feeds	2.3.4; H2c.

Li, H., Ahmed, S. & Chevapatrakul, T. (2016) Volatility spillovers across European stock markets around the Brexit referendum. Nottingham CFCM WP 16/06.	
Source basis	Full text read in this session — facts checked against the paper
Question	How did volatility spill over between major European markets around the referendum?
Data	15-minute intraday data; France, Germany, Switzerland, UK; 4 Jan – 30 Sep 2016.
Method	Diebold–Yilmaz style total, directional, net and pairwise spillovers.
Key findings	• Strong interdependence among the four markets. • Over the whole period, France and Germany were net transmitters; the UK and Switzerland net receivers. • Around referendum day, France and the UK were net transmitters.
Limitations	Four large markets only; no Ireland.
How to use it	Competing evidence for H2d: the UK was a transmitter around the vote but not overall. Justifies watching the euro-core channel in your Brexit check too.
Feeds	2.3.5; H2d.

Li, H. (2020) Volatility spillovers across European stock markets under the uncertainty of Brexit. Economic Modelling, 84, 1–12.	
Source basis	Abstract / official summary read in this session — read the full paper for details
Question	How did Brexit uncertainty change volatility spillovers in Europe?
Data	European stock markets from January 2016 through and after the vote.
Method	Time-varying multivariate GARCH; variance decomposition; volatility impulse responses.
Key findings	• The UK's influence on other markets decreased from the start of the referendum campaign. • The vote raised volatility, with diverse instant effects on co-movement. • Its effect on co-volatility remained substantial and persistent.
Limitations	Market list not confirmed (check whether Ireland is included).
How to use it	Second source for the competing view on H2d.
Feeds	2.3.5; H2d.

Boungou, W. & Yatié, A. (2022) The impact of the Ukraine–Russia war on world stock market returns. Economics Letters, 215, 110516.	
Source basis	Abstract / official summary read in this session — read the full paper for details
Question	How did the invasion affect stock markets worldwide?
Data	Daily returns of 94 countries, 22 Jan – 24 Mar 2022.
Method	Panel regression around the invasion.
Key findings	• Negative relationship between the war and world stock returns. • Largest impact at the onset, especially in the first two weeks after 24 February.
Limitations	Returns, not volatility; short window; no rates channel.
How to use it	Background for your 2022 validation event (H2e). Add the rates channel from the ECB/Fed tightening and Minsky's point on monetary tightening.
Feeds	2.3.5; H2e.

Kearney, C. (1998) The causes of volatility in a small, internationally integrated stock market: Ireland, July 1975–June 1994. Journal of Financial Research, 21(1).	
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Source basis	Abstract / official summary read in this session — read the full paper for details
Question	What drives volatility in the Irish stock market?
Data	Irish market, 1975–1994.
Method	(Read the paper.)
Key findings	• Title confirmed; treats Ireland as a small, internationally integrated market. Read for findings.
Limitations	Old sample; pre-euro.
How to use it	Historical Irish evidence that foreign factors matter — a precedent for your design.
Feeds	2.3.1; H2d.

Twomey, C. (1998) Identifying the source of mean and volatility spillovers in Irish equities: a multivariate GARCH analysis. The Economic and Social Review.	
Source basis	Abstract / official summary read in this session — read the full paper for details
Question	Where do Irish equity return and volatility spillovers come from?
Data	Ten largest Irish stocks, the Irish and UK markets.
Method	Multivariate VAR-GARCH.
Key findings	• Significant mean and volatility spillovers from the UK market to individual Irish stocks.
Limitations	Pre-euro, 1990s data; confirm volume and pages.
How to use it	Early Irish evidence: the basis for expecting a UK channel (H2d).
Feeds	2.3.1; H2d.

D3. Section 2.4 — How shocks travel

Forbes, K. & Rigobon, R. (2002) No contagion, only interdependence: measuring stock market co-movements. Journal of Finance, 57(5), 2223–2261 (NBER WP 7267).	
Source basis	Full text read in this session — facts checked against the paper
Question	Does correlation between markets really rise in crises, or is it a statistical artefact?
Data	Stock markets in the 1997 East Asian crisis, 1994 Mexican peso crisis and 1987 US crash.
Method	Correlation tests, with and without a volatility adjustment.
Key findings	• The standard correlation is biased upward when volatility rises. • Unadjusted tests find contagion in several countries; adjusted tests find virtually none. • High co-movement in these crises was a continuation of strong existing links (interdependence). • The adjusted coefficient can still be biased by endogeneity or common unobserved shocks.
Limitations	Assumes no endogeneity/omitted variables; only three crises.
How to use it	Defines your vocabulary and the key caution: rising crisis-period attributions may reflect volatility, not new channels.
Feeds	2.4; H2 interpretation; Limitations.

Engle, R., Ito, T. & Lin, W.-L. (1990) Meteor showers or heat waves? Heteroskedastic intra-daily volatility in the foreign exchange market. Econometrica, 58(3), 525–542 (NBER WP 2609).	
Source basis	Full text read in this session — facts checked against the paper
Question	Does volatility stay in one market (heat wave) or travel across markets (meteor shower)?
Data	Yen/dollar exchange rate across market segments (Tokyo, Europe, New York).
Method	ARCH models across intra-daily segments; volatility VAR.
Key findings	• The pure heat-wave hypothesis is rejected: volatility spills over from one market to the next. • News in one market affects the path of volatility in others.
Limitations	Foreign exchange, not equities; 1980s data.
How to use it	Perfect metaphor for your channel groups: domestic = heat wave, foreign channels = meteor showers.
Feeds	2.4; RQ2 design.

Hamao, Y., Masulis, R. & Ng, V. (1990) Correlations in price changes and volatility across international stock markets. Review of Financial Studies, 3(2), 281–307.	
Source basis	Abstract / official summary read in this session — read the full paper for details
Question	How do price changes and volatility spill over between Tokyo, London and New York?
Data	Daily open and close prices of major indices.
Method	ARCH-family models.
Key findings	• Volatility spillovers from New York to Tokyo, London to Tokyo, and New York to London.
Limitations	Three markets; late-1980s data.
How to use it	Justifies your time-zone design (New York closes after Dublin; use day-t US data for day t+1).
Feeds	2.4; Chapter 3 alignment.

King, M. & Wadhwani, S. (1990) Transmission of volatility between stock markets. Review of Financial Studies, 3(1), 5–33.	
Source basis	Standard textbook knowledge, not re-read this session — confirm every detail before citing
Question	Why did correlations between markets rise around the 1987 crash?
Data	New York, London, Tokyo around October 1987.
Method	Contagion model with information inference.
Key findings	• Traders infer information from price moves in other markets, so mistakes spread; correlations rise with volatility.
Limitations	Single event.
How to use it	Early theory of contagion as information spillover.
Feeds	2.4.

Kearney, C. (2000) The determination and international transmission of stock market volatility. Global Finance Journal, 11, 31–52.	
Source basis	Abstract / official summary read in this session — read the full paper for details
Question	What drives stock-market volatility and how is it transmitted internationally?
Data	Monthly data, 1973–1994: Britain, France, Germany, Japan, US (returns, rates, FX, inflation, output).
Method	Portfolios, cointegration, conditional volatility transmission.
Key findings	• Documents cointegrating relationships between world, European and Japanese/US equity portfolios and transmission of conditional volatility between them.
Limitations	Monthly, pre-2000.
How to use it	Low-frequency transmission evidence by an Irish-based author.
Feeds	2.4.

Diebold, F. & Yilmaz, K. (2009) Measuring financial asset return and volatility spillovers, with application to global equity markets. Economic Journal, 119(534), 158–171 (NBER WP 13811).	
Source basis	Full text read in this session — facts checked against the paper
Question	How can interdependence be measured simply, in calm and crisis periods?
Data	19 global equity markets, early 1990s – 2007.
Method	Spillover index from VAR forecast-error variance decompositions.
Key findings	• Separate measures for return and volatility spillovers. • Return spillovers: gently increasing trend, no bursts. • Volatility spillovers: no trend but clear bursts in crises.
Limitations	Order-dependent in this version (fixed by the generalised version in 2012).
How to use it	Evidence that volatility transmission bursts in crises (supports expecting foreign shares to rise); name it as the econometric validation you will add in the PhD.
Feeds	2.4; H2; Future research.

Sahiner, M. (2023) Volatility spillovers and contagion during major crises: an early warning approach based on a deep learning model. Computational Economics.	
Source basis	Abstract / official summary read in this session — read the full paper for details
Question	How is volatility transmitted in major crashes, and can deep learning give early warning?
Data	International stock markets, 3 Jul 1997 – 9 Mar 2021.
Method	DCC, GARCH-BEKK, Diebold–Yilmaz index, plus a deep-learning early-warning model.
Key findings	• External shocks have long-lasting domestic effects through contagion in crises. • Asian financial stress was transmitted mainly via Hong Kong.
Limitations	No explanation-based channel attribution; not Ireland; no cross-crisis generalisation test.
How to use it	Closest prior work — cite it and state exactly how you differ (see Blueprint 5.12).
Feeds	2.4; 2.7 gap.

Pericoli & Sbracia (2003) J. Economic Surveys 17(4); Dungey et al. (2005) Quantitative Finance 5(1); Bekaert, Harvey & Ng (2005) J. Business 78(1).	
Source basis	Standard textbook knowledge, not re-read this session — confirm every detail before citing
Question	How should contagion be defined and measured?
Data	Reviews and a factor-model study.
Method	Surveys; factor model of integration.
Key findings	• Several competing definitions of contagion exist (Pericoli & Sbracia). • Many contagion tests fit into a common latent-factor framework (Dungey et al.).

	• Contagion measured as excess correlation of factor-model residuals (Bekaert, Harvey & Ng).
Limitations	Confirm details before citing.
How to use it	Use for definitions and method overview in 2.4.
Feeds	2.4.

D4. Section 2.5 — Forecasting volatility

Engle, R. (1982) <i>Econometrica</i> 50(4); Bollerslev, T. (1986) <i>Journal of Econometrics</i> 31(3).	
Source basis	Standard textbook knowledge, not re-read this session — confirm every detail before citing
Question	How can time-varying variance be modelled?
Data	UK inflation (Engle); general (Bollerslev).
Method	ARCH; GARCH.
Key findings	• ARCH: variance depends on past squared shocks. • GARCH: adds past variance, giving a parsimonious model of persistent volatility.
Limitations	Symmetric response to good and bad news (GJR-GARCH adds the leverage effect).
How to use it	Foundations of your first benchmark.
Feeds	2.5.

Hansen, P. & Lunde, A. (2005) A forecast comparison of volatility models: does anything beat a GARCH(1,1)? <i>Journal of Applied Econometrics</i>, 20(7), 873–889.	
Source basis	Standard textbook knowledge, not re-read this session — confirm every detail before citing
Question	Do more complex volatility models beat GARCH(1,1)?
Data	DM–USD exchange rate and IBM stock returns.
Method	330 ARCH-type models compared with realised variance and a superior-predictive-ability test.
Key findings	• Exchange rate: no evidence that GARCH(1,1) is beaten. • IBM stock: GARCH(1,1) is inferior to models with a leverage effect.
Limitations	Two series only.
How to use it	Sets the bar: GARCH(1,1) is a serious benchmark; for equities, a leverage version (GJR) could be a robustness check.
Feeds	2.5; H1.

Andersen, T. & Bollerslev, T. (1998) <i>International Economic Review</i> 39(4); Andersen, Bollerslev, Diebold & Labys (2003) <i>Econometrica</i> 71(2).	
Source basis	Standard textbook knowledge, not re-read this session — confirm every detail before citing
Question	Are volatility models really poor, or is the yardstick poor? How should realised volatility be modelled?
Data	Foreign exchange, high-frequency data.
Method	Realised volatility from intraday returns; long-memory models of log RV.
Key findings	• GARCH looks poor when judged against noisy squared daily returns, but good against realised volatility. • Modelling log realised volatility directly gives strong forecasts.
Limitations	Rely on intraday data (you use daily closes).
How to use it	Why you use 5-day realised variance (less noise) and model it in logs.
Feeds	2.5; Chapter 3.

Corsi, F. (2009) A simple approximate long-memory model of realized volatility. <i>Journal of Financial Econometrics</i>, 7(2), 174–196.	
Source basis	Standard textbook knowledge, not re-read this session — confirm every detail before citing
Question	Can a simple model capture volatility's long memory?
Data	High-frequency financial data.
Method	HAR-RV: OLS on daily, weekly and monthly realised volatility.
Key findings	• Three components reproduce long memory and fat tails. • Forecasts well despite its simplicity; now the standard benchmark.
Limitations	Linear; designed for intraday-based RV.
How to use it	Your second benchmark and the natural competitor for an LSTM (both use a history of volatility).
Feeds	2.5; H1.

Patton, A. (2011) Volatility forecast comparison using imperfect volatility proxies. <i>Journal of Econometrics</i>, 160(1), 246–256.	
Source basis	Full text read in this session — facts checked against the paper
Question	Can forecast comparisons be trusted when true volatility is unobserved?

Data	Theory; application to IBM volatility.
Method	Analytical results on loss functions.
Key findings	- Noisy proxies can make a worse forecast look better under some loss functions — affecting test size, not just power. - Less noisy proxies (range, realised volatility) reduce but do not remove the distortion. - Derives conditions for robust loss functions; a family that nests MSE and QLIKE.
Limitations	Theory-focused.
How to use it	Justifies reporting MSE and QLIKE, with QLIKE primary.
Feeds	2.5; Chapter 3; H1.

Diebold, F. & Mariano, R. (1995) JBES 13(3); Harvey, D., Leybourne, S. & Newbold, P. (1997) IJF 13(2).	
Source basis	Standard textbook knowledge, not re-read this session — confirm every detail before citing
Question	How can two forecasts be compared formally?
Data	General.
Method	Test on the mean loss differential; small-sample fix.
Key findings	- DM: tests whether the average loss difference is zero, robust to autocorrelation. - HLN: corrects the DM statistic for small samples and multi-step forecasts; use t critical values.
Limitations	Designed for non-nested comparisons; see literature on nested models.
How to use it	Your RQ1 test.
Feeds	2.5; Chapter 3.

Hochreiter, S. & Schmidhuber, J. (1997) Long short-term memory. Neural Computation, 9(8), 1735–1780.	
Source basis	Standard textbook knowledge, not re-read this session — confirm every detail before citing
Question	How can recurrent networks learn long time lags?
Data	Artificial benchmark tasks.
Method	LSTM architecture.
Key findings	- Constant error carousel keeps gradients from vanishing; gates control access. - Learns dependencies over very long lags where plain RNNs fail.
Limitations	Original version had no forget gate (added by Gers et al., 2000).
How to use it	Cite for the LSTM itself in 2.5 and Chapter 3.
Feeds	2.5.

Fischer, T. & Krauss, C. (2018) Deep learning with LSTM networks for financial market predictions. European Journal of Operational Research, 270(2), 654–669.	
Source basis	Standard textbook knowledge, not re-read this session — confirm every detail before citing
Question	Can LSTMs predict stock movements better than other methods?
Data	S&P 500 constituents, 1992–2015.
Method	LSTM vs random forest, deep feed-forward network, logistic regression.
Key findings	- LSTM outperformed the other methods on directional prediction. - Returns declined in later years, consistent with the edge being competed away.
Limitations	Returns direction, not volatility; confirm figures.
How to use it	Evidence that LSTMs capture structure in financial data; contrast with EMH.
Feeds	2.5.

Kim, H. & Won, C. (2018) Forecasting the volatility of stock price index: a hybrid model integrating LSTM with multiple GARCH-type models. Expert Systems with Applications, 103, 25–37.	
Source basis	Standard textbook knowledge, not re-read this session — confirm every detail before citing
Question	Does combining LSTM with GARCH-type forecasts improve volatility prediction?
Data	KOSPI 200 index (Korea).
Method	LSTM fed with GARCH-family outputs (hybrid).
Key findings	Hybrid models reduced forecast errors relative to single models.
Limitations	One market; confirm details.
How to use it	Shows LSTM–GARCH complementarity; a PhD extension idea.
Feeds	2.5.

Bucci, A. (2020) Realized volatility forecasting with neural networks. Journal of Financial Econometrics, 18(3), 502–531.	
Source basis	Full text read in this session — facts checked against the paper
Question	Do neural networks forecast realised volatility better than econometric models?
Data	Monthly S&P 500, Feb 1950 – Dec 2017 (815 observations); log RV from daily squared returns; macro and

	financial predictors.
Method	FNN, Elman, Jordan, LSTM and NARX networks vs ARFIMA(X) and LSTAR(X); out-of-sample 1997–2017 and a 2007–09 subsample; MSE and QLIKE; Model Confidence Set.
Key findings	• Recurrent networks outperform traditional econometric methods. • LSTM and NARX — the long-memory networks — are best, and consistently beat FNN, Elman and Jordan. • Their advantage also holds in the highly volatile 2007–09 period.
Limitations	Monthly data (yours is daily/weekly); US only; no explanation of predictions.
How to use it	Main evidence for H1, especially for crises. Note the frequency difference.
Feeds	2.5; H1.

Christensen, K., Siggaard, M. & Veliyev, B. (2023) A machine learning approach to volatility forecasting. Journal of Financial Econometrics, 21(5), 1680–1727.	
Source basis	Full text read in this session — facts checked against the paper
Question	How accurate is machine learning at forecasting realised variance compared with HAR models?
Data	DJIA constituents, 29 Jan 2001 – 31 Dec 2017 (4,257 trading days).
Method	Regularised regression, trees, random forest, gradient boosting, neural networks vs a suite of HAR models; minimal tuning; 1-day, 1-week, 1-month horizons; accumulated local effects (ALE) for importance.
Key findings	• ML is competitive and beats the HAR family even with only HAR's own lags as inputs. • Gains are minor for next-day forecasts but larger at weekly and monthly horizons (better long-memory capture). • Random forest and neural networks are best; ML finds extra information in additional predictors. • Models agree on the most important predictors but disagree on their ranking. • Lower losses in a Value-at-Risk application, less significant.
Limitations	US large caps; intraday RV.
How to use it	Supports H1 and your weekly (5-day) target. The ranking-disagreement finding motivates your RQ4 stability test.
Feeds	2.5, 2.6; H1; H4.

Bergmeir, C. & Benítez, J. (2012) On the use of cross-validation for time series predictor evaluation. Information Sciences, 191, 192–213.	
Source basis	Standard textbook knowledge, not re-read this session — confirm every detail before citing
Question	How should forecasting models be validated on time series?
Data	Simulations and real series.
Method	Comparison of validation schemes.
Key findings	• Random shuffling breaks time order; blocked/forward schemes are safer.
Limitations	Confirm details.
How to use it	Justifies walk-forward (expanding-window) evaluation.
Feeds	2.5; Chapter 3.

D5. Section 2.6 — Explaining ML forecasts

Lundberg, S. & Lee, S.-I. (2017) A unified approach to interpreting model predictions. NeurIPS 30.	
Source basis	Full text read in this session — facts checked against the paper
Question	Can the many explanation methods be unified on a sound theoretical basis?
Data	Theory plus experiments.
Method	Additive feature attribution framework based on Shapley values.
Key findings	• Defines a class of additive attribution methods that includes six earlier methods. • Shows a unique solution with local accuracy, missingness and consistency: SHAP values. • Estimators: Kernel SHAP (model-agnostic), Deep SHAP (DeepLIFT + Shapley), and others. • Accuracy vs interpretability tension motivates the work.
Limitations	Exact Shapley values are expensive; estimators make assumptions (e.g., feature independence).
How to use it	Cite for SHAP's definition and properties.
Feeds	2.6; Chapter 3.

Sundararajan, M., Taly, A. & Yan, Q. (2017) Axiomatic attribution for deep networks. ICML.	
Source basis	Full text read in this session — facts checked against the paper
Question	What properties should attributions for deep networks satisfy?
Data	Image, text and chemistry models.
Method	Axioms and the Integrated Gradients method.
Key findings	• Two axioms — Sensitivity and Implementation Invariance — are violated by most earlier methods. • Integrated Gradients satisfies them, needs only gradient calls and no change to the network.

	• Does not explain interactions between features.
Limitations	Result depends on the baseline chosen.
How to use it	Theoretical basis for gradient-based SHAP on your LSTM.
Feeds	2.6.

Erion, G. et al. (2021) Improving performance of deep learning models with axiomatic attribution priors and expected gradients. *Nature Machine Intelligence*, 3, 620–631.

Source basis	Full text read in this session — facts checked against the paper
Question	Can a fast attribution method satisfy the axioms, and can attributions improve training?
Data	Images, genomics, health data.
Method	Expected gradients (IG averaged over baselines from the data).
Key findings	• Expected gradients satisfies key axioms such as completeness and implementation invariance. • Using it in training (attribution priors) improves performance.
Limitations	Main focus is training, not explanation per se.
How to use it	Justifies GradientShap with a background sample of real training windows.
Feeds	2.6; Chapter 3.

Aas, K., Jullum, M. & Løland, A. (2021) Explaining individual predictions when features are dependent. *Artificial Intelligence*, 298, 103502.

Source basis	Full text read in this session — facts checked against the paper
Question	What goes wrong with Kernel SHAP when features are correlated?
Data	Simulated linear and non-linear models.
Method	Extensions of Kernel SHAP for dependent features.
Key findings	• Kernel SHAP assumes independent features, which can give very wrong explanations — even for linear models. • Their dependence-aware versions approximate true Shapley values more accurately.
Limitations	Computationally heavier.
How to use it	Justifies channel grouping and the Nasdaq-excess-return design.
Feeds	2.6; H4.

Kumar, I., Venkatasubramanian, S., Scheidegger, C. & Friedler, S. (2020) Problems with Shapley-value-based explanations as feature importance measures. *ICML*.

Source basis	Full text read in this session — facts checked against the paper
Question	Are Shapley values a sound general answer to ‘feature importance’?
Data	Theoretical analysis.
Method	Mathematical and human-centred critique.
Key findings	• Mathematical problems arise when Shapley values are used for feature importance. • Fixing them requires extra complexity such as causal reasoning. • Shapley values do not naturally meet human explanation goals; approaches should fit specific use cases.
Limitations	Critique, not an alternative method.
How to use it	Your main ‘limits of SHAP’ source; supports the interpretation rule.
Feeds	2.6; Limitations.

Rudin, C. (2019) Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. *Nature Machine Intelligence*, 1, 206–215.

Source basis	Full text read in this session — facts checked against the paper
Question	Should we explain black boxes or build interpretable models?
Data	Examples from criminal justice, healthcare and more.
Method	Position paper.
Key findings	• Explaining black boxes in high-stakes settings can perpetuate bad practice and cause harm. • Inherently interpretable models should be preferred where possible.
Limitations	Argumentative.
How to use it	The counter-argument to engage with. Your answer: interpretable benchmarks (HAR, GARCH) are included, and SHAP is used for research diagnosis, not automated decisions.
Feeds	2.6; Discussion.

Ahmed, A. (2026) Explainable deep learning for financial volatility forecasting: an LSTM-Attention-SHAP framework with comprehensive validation. *SSRN preprint 6301119*.

Source basis	Abstract / official summary read in this session — read the full paper for details
Question	Can an LSTM with attention and SHAP forecast volatility accurately and transparently?

Data	Five asset classes, Jan 2018 – Dec 2024.
Method	LSTM + attention + SHAP.
Key findings	• Reported 28.6% lower RMSE than GARCH(1,1) (per abstract summary — verify). • SHAP ranks a geopolitical risk index first for extreme volatility, then VIX (per abstract summary — verify).
Limitations	Preprint, not peer-reviewed; RMSE only; descriptive use of SHAP.
How to use it	Closest explainable-volatility work: SHAP used descriptively, not as a test across crises. Cite as a preprint.
Feeds	2.6; 2.7 gap.

E. From notes to your own paragraph

Write each paragraph from several cards, never from one. Plan it as five parts before you write a sentence:

Part	What it does	Planning example (bullet notes, not text to copy)
1 Point	Your claim, in your words	GFC reached Ireland through both a US channel and a domestic one
2 Evidence	Two or more sources that support it	Bekaert et al. — little US contagion, strong domestic contagion; Kelly — 7×/11× lending boom; Regling & Watson — home-grown
3 Contrast	A source that complicates it	Reinhart & Rogoff — crises usually start in financial centres, so the US trigger still matters
4 Critique	Limits of the evidence	Bekaert's factor-model definition; the official reports are not tests
5 So what	Link to your study	Expect both US and domestic channels to rise in the GFC window (H2a)

- Group studies by what they found or how they measured it, not by author.
- Every paragraph ends with its 'so what' for your research questions.
- Cite the original paper, never this handbook, and check page numbers for any figure you quote.
- Use at most a few words of direct quotation, in quotation marks with a page number, and only when the exact wording matters.

F. Where to read the originals

Open-access versions used for this handbook. For journal versions, log in through the DBS library.

Source	Link
Minsky (1992)	levyinstitute.org/pubs/wp74.pdf
Reinhart & Rogoff (NBER WP 13882)	nber.org/papers/w13882
Acharya, Drechsler & Schnabl	pages.stern.nyu.edu/~pschnabl/research/AcharyaDrechslerSchnabl_final.pdf
Bekaert et al. (ECB WP 1381)	ecb.europa.eu/pub/pdf/scpwp/scpwp1381.pdf
Baker et al. (NBER WP 26945)	nber.org/papers/w26945
Forbes & Rigobon (NBER WP 7267)	nber.org/papers/w7267
Engle, Ito & Lin (NBER WP 2609)	nber.org/papers/w2609
Diebold & Yilmaz 2009 (NBER WP 13811)	nber.org/papers/w13811
Patton (2011)	public.econ.duke.edu/~ap172/Patton_vol_proxies_JoE_2011.pdf
Bucci (MPRA 95443)	mpra.ub.uni-muenchen.de/95443/
Christensen et al.	arxiv.org/abs/2601.13014
Akhtaruzzaman et al.	ideas.repec.org/a/eee/finlet/v38y2021ics1544612320305754.html
Li, Ahmed & Chevapatrakul (CFCM 16/06)	nottingham.ac.uk/cfcm/news/papers/1606.aspx
Lundberg & Lee	arxiv.org/abs/1705.07874
Sundararajan et al.	arxiv.org/abs/1703.01365
Erion et al.	arxiv.org/abs/1906.10670
Aas et al.	arxiv.org/abs/1903.10464
Kumar et al.	arxiv.org/abs/2002.11097
Rudin	arxiv.org/abs/1811.10154
Kelly (2009)	ideas.repec.org/p/ucn/wpaper/200932.html
Whelan (2014)	ideas.repec.org/p/ucn/wpaper/201306.html
Lane (2012), Brunnermeier (2009)	aeaweb.org (JEP is free to read)
Corbet & Twomey (2015)	ideas.repec.org/a/prg/jnlefa/v2015y2015i3id143p15-34.html
Sahiner (2023)	doi.org/10.1007/s10614-023-10412-4
Li (2020)	ideas.repec.org/a/eee/ecmode/v84y2020icp1-12.html

